Practical no. 14

Aim – File Upload and Download Automation
Develop a script to automate file upload and download functionality. Validate successful upload and verify file existence after download.

Tool: Selenium WebDriver + Java Robot Class + Eclipse

Theory –

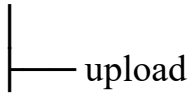
- Selenium: Open-source tool for automating web applications.
- WebDriver: Controls and interacts with web browsers.
- Locator: Identifies web elements, e.g. id, name, xpath, cssSelector.
- Robot Class: Java class used to perform keyboard and mouse actions at OS level.
- Robot Package: java.awt.Robot
- Selenium is used for: Web-page automation.
- Robot is used for: File dialogs, keyboard shortcuts, mouse actions, etc.
- Eclipse: IDE used to write, run, and debug Java programs.

STEP 1 — Check Java and Eclipse
Before creating the project, make sure you have:

1. **JDK**
2. **Eclipse IDE for Java Developers**
3. **Google Chrome**
4. **Selenium Java libraries**

Check Java
Open **Command Prompt** and type:
java -version
You should see something like:
java version "17..." or
java version "21..."
If Java is installed, continue.

STEP 2 — Create Upload Folder
Now we need a file that our program will upload.
Go to Windows File Explorer.
Create: D:\STQA\FileUploadDownload
Inside it create: upload
and: download
So you have:D:\STQA\FileUploadDownload



└── download

STEP 3 — Create Student File
Open: D:\STQA\FileUploadDownload\upload
Right-click → **New** → **Text Document**
Rename it: student.txt
Open the file and type:
This is a Selenium file upload test. Student Record
Save it.

STEP 4 — Open Eclipse
Open **Eclipse**.
You may see: Eclipse IDE Workspace
Click: **Launch**

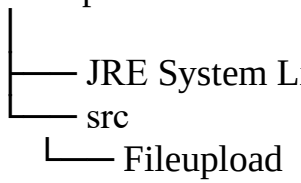
STEP 5 — Create Java Project
From the Eclipse menu:
File
↓
New
↓
Java Project
A window will appear.
Enter: Project name: FileUploadDownload
Click:**Finish**

STEP 6 — Check Your Project
On the left side, in **Package Explorer**, you should see: FileUploadDownload
Click the arrow beside it. You will see something similar to:
FileUploadDownload
├── JRE System Library
└── src
src means **Source**.
Your Java program will be created inside src.

STEP 7 — Create a Package
Right-click: src
Select: New
↓
Package
A window will appear.
Enter: Fileupload

Click: **Finish**

Now your project looks like:
FileUploadDownload



STEP 8 — Create Java Class
Right-click the package: Fileupload
Select: New
↓
Class
A window will appear.
Enter :- Name: FileUploadDownload
Make sure this option is selected:
☑ public static void main(String[] args)
Then click: **Finish**

STEP 9 — Your Java File
You will now see:
FileUploadDownload.java
and Eclipse will create something like:

STEP 10 — Add Selenium Libraries
Now we need to tell Eclipse where Selenium is.
You should have Selenium JAR files.
In Eclipse Right-click: FileUploadDownload
Select: Build Path
↓
Configure Build Path
A window will open.
Select: Libraries
Then: Classpath
↓
Add External JARs...
Select the Selenium JAR files.
D:\STQA\selenium-java-4.48.0
Select all jar files and click: **Apply and Close** is:
D:\STQA\FileUploadDownload\upload\student.txt

This exact path will be used in the Java program.

— Why Are We Using Robot Class?
When Selenium clicks:
driver.findElement(By.id("file-upload")).click();
the Windows file selection window opens.
The normal Selenium commands don't directly control that native Windows dialog.
So we use: Robot robot = new Robot();
Robot can simulate keyboard actions.

STEP 11 — Complete Code

After completing all the above steps, your **complete Eclipse program** should look like this:

```
package Fileupload;
import java.awt.Robot;
import java.awt.Toolkit;
import java.awt.datatransfer.StringSelection;
import java.awt.event.KeyEvent;
import java.io.File;
import java.time.Duration;
import java.util.HashMap;
import java.util.Map;
import org.openqa.selenium.By;
import org.openqa.selenium.JavascriptExecutor;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.chrome.ChromeDriver;
import org.openqa.selenium.chrome.ChromeOptions;
import org.openqa.selenium.support.ui.ExpectedConditions;
import org.openqa.selenium.support.ui.WebDriverWait;
public class FileUploadDownload {
    public static void main(String[] args) throws
Exception {
        // FILE PATHS
        String uploadFilePath =

"D:\\STQA\\FileUploadDownload\\upload\\student.txt";
        String downloadFolderPath =
                "D:\\STQA\\FileUploadDownload\\download";
        // CHECK UPLOAD FILE
        File uploadFile =
```

```
        new File(uploadFilePath);
if (!uploadFile.exists()) {
    System.out.println(
            "ERROR: Upload file not found!");
    System.out.println(
            "File path: " + uploadFilePath);
    return;
}
System.out.println(
        "Upload file found:");
System.out.println(
        uploadFilePath);
// CREATE DOWNLOAD FOLDER
File downloadFolder =
        new File(downloadFolderPath);
if (!downloadFolder.exists()) {
    downloadFolder.mkdirs();
    System.out.println(
            "Download folder created.");
}
// CHROME OPTIONS
ChromeOptions options =
        new ChromeOptions();
Map<String, Object> prefs =
        new HashMap<>();
prefs.put(
        "download.default_directory",
        downloadFolderPath);
prefs.put(
        "download.prompt_for_download",
        false);
prefs.put(
        "download.directory_upgrade",
        true);
options.setExperimentalOption(
        "prefs",
        prefs);
// START CHROME
WebDriver driver =
        new ChromeDriver(options);
driver.manage()
        .window()
```

```
                .maximize();
        try {
            // FILE UPLOAD
            System.out.println();
            System.out.println(
                    "Opening File Upload page...");
            driver.get(
                    "https://the-
internet.herokuapp.com/upload");
            Thread.sleep(2000);
            // FIND CHOOSE FILE ELEMENT
            WebElement uploadElement =
                    driver.findElement(
                            By.id("file-upload"));
            // OPEN WINDOWS FILE DIALOG
            JavascriptExecutor js =
                    (JavascriptExecutor) driver;
            js.executeScript(
                    "arguments[0].click();",
                    uploadElement);
            Thread.sleep(2000);
            // ROBOT CLASS
            Robot robot =
                    new Robot();
            robot.setAutoDelay(100);
            // COPY FILE PATH
            StringSelection selection =
                    new StringSelection(
                            uploadFilePath);
            Toolkit.getDefaultToolkit()
                    .getSystemClipboard()
                    .setContents(
                            selection,
                            null);

            Thread.sleep(1000);
            // PASTE FILE PATH
            robot.keyPress(
                    KeyEvent.VK_CONTROL);
            robot.keyPress(
                    KeyEvent.VK_V);
            robot.keyRelease(
```

```
        KeyEvent.VK_V);
robot.keyRelease(
        KeyEvent.VK_CONTROL);
Thread.sleep(1000);
// PRESS ENTER
robot.keyPress(
        KeyEvent.VK_ENTER);
robot.keyRelease(
        KeyEvent.VK_ENTER);
Thread.sleep(3000);
System.out.println(
        "File selected using Robot Class.");
// CLICK UPLOAD
System.out.println();

System.out.println(
        "Clicking Upload button...");
driver.findElement(
        By.id("file-submit"))
        .click();
// WAIT FOR UPLOAD RESULT
WebDriverWait wait =
        new WebDriverWait(
                driver,
                Duration.ofSeconds(10));
wait.until(
        ExpectedConditions
                .presenceOfElementLocated(
                        By.tagName("body")));
Thread.sleep(2000);
// GET UPLOAD RESULT
String uploadResult =
        driver.findElement(
                By.tagName("body"))
                .getText();
System.out.println();
System.out.println(
        "Upload Result:");
System.out.println(
        uploadResult);
// VERIFY UPLOAD
if (uploadResult.contains(
```

```
                    "File Uploaded!")) {
                System.out.println();
                System.out.println(
                        "File Upload Successful");
            } else if (uploadResult.contains(
                    "Internal Server Error")) {
                System.out.println();
                System.out.println(
                        "File Upload Failed");
                System.out.println(
                        "Server returned Internal Server
Error.");
            } else {
                System.out.println();
                System.out.println(
                        "File Upload Failed");
            }
            // FILE DOWNLOAD
            System.out.println();
            System.out.println(
                    "Opening File Download page...");
            driver.get(
                    "https://the-
internet.herokuapp.com/download");
            Thread.sleep(2000);
            // COUNT FILES BEFORE DOWNLOAD
            File[] filesBefore =
                    downloadFolder.listFiles();
            int filesBeforeCount =
                    filesBefore == null
                            ? 0
                            : filesBefore.length;
            System.out.println(
                    "Files before download: "
                            + filesBeforeCount);
            // CLICK DOWNLOAD LINK
            driver.findElement(
                    By.cssSelector("#content a"))
                    .click();
            System.out.println(
                    "Download started...");
            // WAIT FOR DOWNLOAD
```

```
            File downloadedFile =
                    null;
            boolean downloadSuccessful =
                    false;
            for (int i = 0; i < 30; i++) {
                Thread.sleep(1000);
                File[] files =
                        downloadFolder.listFiles();
                if (files != null) {
                    for (File file : files) {
                        // Ignore Chrome temporary files
                        if (file.getName()
                                .endsWith(".crdownload"))
{
                            continue;
                        }
                       // Check whether this is a new file
                        if (file.lastModified()
                                >
System.currentTimeMillis()
                                - 30000) {
                            downloadedFile =
                                    file;
                            downloadSuccessful =
                                    true;
                            break;
                        }
                    }
                }
                if (downloadSuccessful) {
                    break;
                }
            }
            // VERIFY DOWNLOAD
            System.out.println();
            if (downloadSuccessful
                    && downloadedFile != null) {
                System.out.println(
                        "File Download Successful");
                System.out.println(
                        "Downloaded File: "
                                + downloadedFile
```

```
                                .getName());
                    System.out.println(
                            "Downloaded File Path: "
                                    + downloadedFile
                                    .getAbsolutePath());
                } else {
                    System.out.println(
                            "File Download Failed");
                }
            } finally {
                // CLOSE BROWSER
                driver.quit();

                System.out.println();

                System.out.println(
                        "Browser closed.");
            }
        }
}
```

STEP 12 — Run in Eclipse
Save the program:
Ctrl + S
Then:
Right-click FileUploadDownload.java
→ Run As
→ Java Application

STEP 13 - Observe the Execution
You should see:
Browser opens
Chrome
Upload page opens
File Uploader
Choose File
Upload
File dialog opens
Robot enters:
D:\STQA\FileUploadDownload\upload\student.txt
Upload happens

Website displays:
File Uploaded!
Eclipse Console
File Upload Successful
Download page opens
A file is downloaded.
Download folder
Check:
D:\STQA\FileUploadDownload\download
You should find a file there.
Eclipse Console
File Download Successful
Downloaded File: some-file.txt

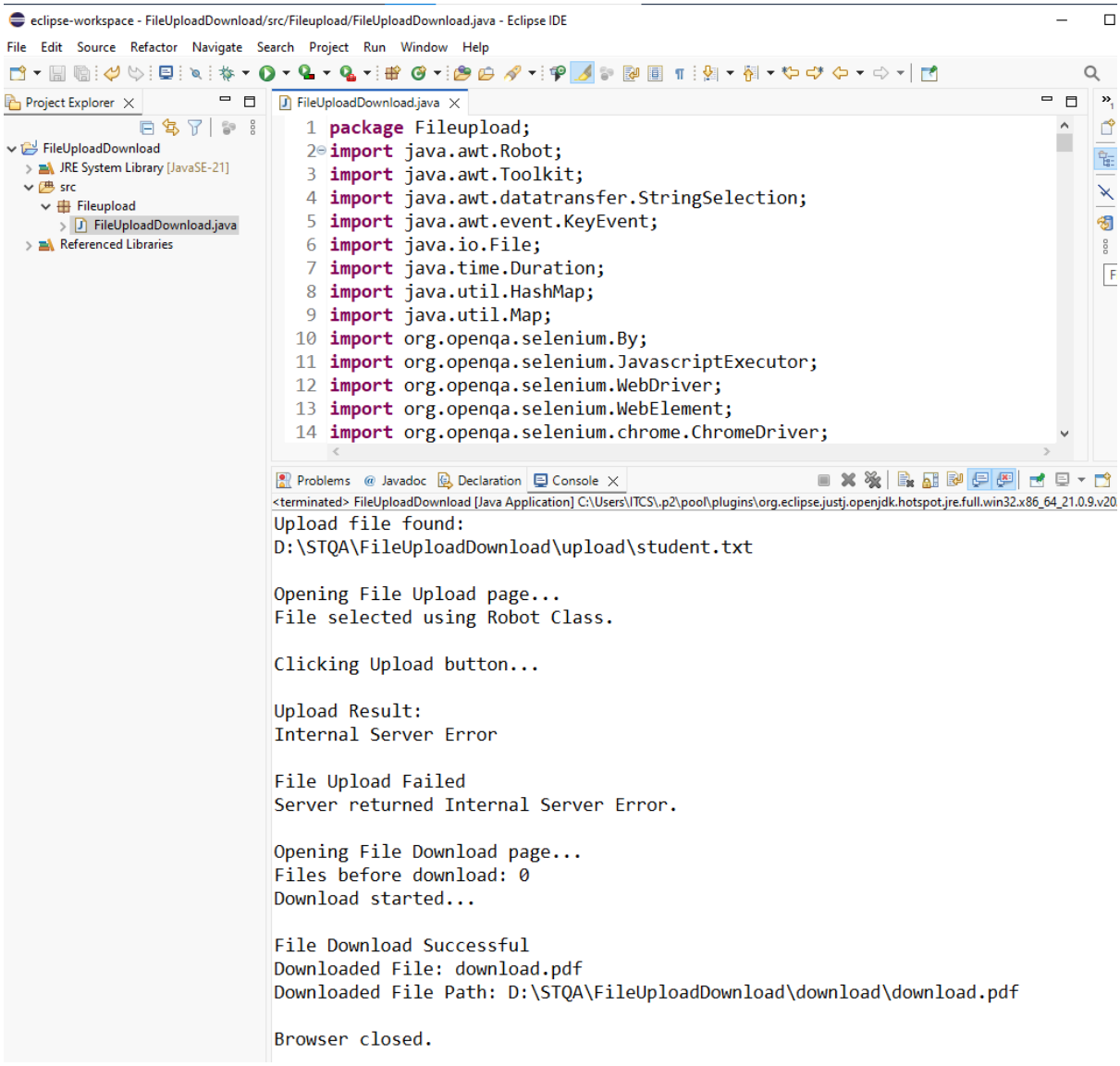
STEP 14 — Final Folder Structure

After execution:
D:\STQA\FileUploadDownload

```
├── upload
│   └── student.txt
│
└── download
    └── downloaded-file.txt
```

Your Eclipse project separately contains:
FileUploadDownload

```
├── JRE System Library
├── Referenced Libraries
│   └── Selenium JAR files
│
└── src
    └── fileupload
        └── FileUploadDownload.java
```


eclipse-workspace - FileUploadDownload/src/Fileupload/FileUploadDownload.java - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help
Project Explorer
FileUploadDownload
JRE System Library [JavaSE-21]
src
Fileupload
FileUploadDownload.java
Referenced Libraries
FileUploadDownload.java
1 package Fileupload;
2 import java.awt.Robot;
3 import java.awt.Toolkit;
4 import java.awt.datatransfer.StringSelection;
5 import java.awt.event.KeyEvent;
6 import java.io.File;
7 import java.time.Duration;
8 import java.util.HashMap;
9 import java.util.Map;
10 import org.openqa.selenium.By;
11 import org.openqa.selenium.JavascriptExecutor;
12 import org.openqa.selenium.WebDriver;
13 import org.openqa.selenium.WebElement;
14 import org.openqa.selenium.chrome.ChromeDriver;
Problems Javadoc Declaration Console
<terminated> FileUploadDownload [Java Application] C:\Users\ITCS\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.9.v20
Upload file found:
D:\STQA\FileUploadDownload\upload\student.txt
Opening File Upload page...
File selected using Robot Class.
Clicking Upload button...
Upload Result:
Internal Server Error
File Upload Failed
Server returned Internal Server Error.
Opening File Download page...
Files before download: 0
Download started...
File Download Successful
Downloaded File: download.pdf
Downloaded File Path: D:\STQA\FileUploadDownload\download\download.pdf
Browser closed.